Deterministic Integer 3SUM at Exponent 1.960614

Quartic Switch Certificates and a Lean Word-RAM Proof

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Abstract

This report documents a deterministic integer 3SUM upper bound of $O(n^{1.960614})$ in the precise word-RAM model used by the accompanying Lean development. It extends the pinned `c0ldzy17/three-sum-lean` formalization with local expansion $c = 232$, a rational quartic switch, integer schedules, and revised execution and resource proofs. A uniform entropy inequality follows from two supporting lines; exact rational interval arithmetic certifies every entry of a twelve-term cost ledger below 1.96061388. The complete formal endpoint is `EndStatement.ThreeSum.SolvedInTime (980307/500000 : Rat)`. Archived official verification records for the pinned source report successful Lean and Nanoda kernel checks, statement comparison, and an allowed-axiom audit. This report explains the analytic certificates, the connection to the formal artifact, and the boundaries of the evidence. The full machine proof is supplied as source rather than reproduced here as a standalone paper proof. The deposit is a technical report and proof artifact, not a claim of journal peer review or a general 3SUM lower bound.

Keywords: 3SUM; deterministic algorithms; word RAM; Lean; formal verification; entropy bounds; exact certificates.

1 Statement and contribution

The input consists of n integers. The decision problem asks whether three pairwise distinct positions have values summing to zero; repeated values at different positions are allowed. The result concerns polynomially bounded integer inputs, not arbitrary real inputs.

Theorem 1.1 (Formal endpoint)**.** *For each fixed $\kappa \in \mathbb{N}$, there exist a deterministic word-RAM program P, a word-width coefficient $b \in \mathbb{N}$, and a time function $T : \mathbb{N} \to \mathbb{N}$ with $T(n) = O(n^{980307/500000})$ such that P correctly decides 3SUM for every input with $|a_i| \le n^\kappa$, at every word width $W \ge b(\lfloor \log_2 n \rfloor + 1)$, within $T(n)$ steps.*

The instruction set in the unchanged specification consists of the constant one, addition, subtraction, multiplication, indirect load and store, a branch on a negative signed word, and accept/reject instructions. Each executed instruction costs one step. The statement quantifies one program per fixed magnitude exponent, uniformly over all sufficiently large word widths. The exact rational exponent is $980307/500000 = 1.960614$.

Theorem 1.1 is the type of `Race.bound` in `proof-source/Submission.lean`. Its proof uses `ThreeSum.solvedInTime_980307_500000` in `Submission/ThreeSum/FinalSolver.lean`. The exact operational semantics and quantifiers are in `Submission/EndStatement.lean`. For a rational exponent r, that specification expresses the time bound as $\exists K \in \mathbb{N}\, \forall n \ge 2,\ T(n)^{r.\mathrm{den}} \le K n^{r.\mathrm{num}}$.

The construction inherits the algorithmic framework, reductions, machine model, and substantial proofs from earlier work [1, 2, 3]. The changes documented here are the expansion-232 instantiation, a positive-coefficient quartic certificate, rounded integer schedules, and the machine-level proof changes needed to connect them to the exact endpoint. The baseline is pinned to commit `2b639655e07421b9ac743de9090459bab473a7c0`; the verified extension is pinned to `8c25c0325de5be4286413f39ca0c9278aad33f52` [4]. No exhaustive novelty search or claim to invent the inherited framework is made. The earlier manuscript at exponent 1.999017 concerns a different stage of the analysis and is not relabelled as this result.

2 Entropy functions and an exact quartic switch

Let $H(t) = -t\log_2 t - (1-t)\log_2(1-t)$, continuously extended at 0 and 1. For an integer $c \geq 3$, set

$$E_c = \tfrac{c}{2}H(1/c), \tag{1}$$
$$k_c = 2cH(1/(2c)) - cH(1/c) - 1, \tag{2}$$
$$\gamma_c(t) = cH(1/c) - cH((1-t)/c). \tag{3}$$

The notation k_c denotes a local combinatorial exponent and is distinct from the input-magnitude exponent κ.

Lemma 2.1 (Positive expansion)**.** *For $0 \leq t \leq 1/2$,*

$$\gamma_c(t) = \log_2(c-1)t + \frac{1}{\ln 2}\sum_{j=2}^{\infty}\frac{1+(-1)^j/(c-1)^{j-1}}{j(j-1)}t^j.$$

All coefficients in this expansion are positive.

Proof. Differentiate to obtain

$$\gamma_c'(t) = \frac{\ln(c-1) + \ln(1+t/(c-1)) - \ln(1-t)}{\ln 2}.$$

The logarithm series converge absolutely and uniformly on the interval. Integrating term by term from zero gives the formula. Since $c-1 \geq 2$, the displayed coefficient numerator is strictly positive for every j. □

For $c = 232$, use the polynomial

$$P(t) = 7.85174904t + 0.72447023t^2 + 0.24044466t^3 + 0.12022459t^4. \tag{4}$$

Every decimal here and below is an exact rational. Rational logarithm enclosures verify that each coefficient is below the corresponding coefficient of Lemma 2.1. Thus $P \leq \gamma_{232}$ on $[0, 1/2]$. Positivity of the coefficients gives monotonicity and convexity; the same certificate verifies $P'(1/2) < 9$.

3 A uniform entropy inequality

The switch requires an inequality valid on an entire interval, not just at sampled values. Fix

$$\rho = 0.27905030, \qquad u = 0.17200068185091493,$$

and define $A = -\log_2(1-u)$ and $B = \log_2((1-u)/u)$. Concavity of entropy and convexity of P give

$$H(t) \leq A + Bt, \qquad P(t) \geq P(u) + (t-u)P'(u).$$

The difference

$$L(t) = \rho[1 + P(u) + (t-u)P'(u)] - (A + Bt)$$

is affine. The exact certificate supplies the strict bounds

$$L(0) > 1.486795215 \cdot 10^{-9}, \qquad L(1/2) > 7.676481609 \cdot 10^{-9}.$$

Linear interpolation proves $L(t) > 0$ throughout $[0, 1/2]$, including its endpoints. Consequently,

$$H(t) < \rho(1 + P(t)) \qquad (0 \le t \le 1/2). \tag{5}$$

The interval bounds certify the two endpoint comparisons; the supporting line argument explains why those comparisons imply a continuous theorem.

4 Finite schedules and local expansion

Use the parameters

$$\begin{aligned} e &= 0.203889696, \qquad & b &= -0.062656016, \\ S &= 0.193175570, \qquad & \theta &= 0.040409899, \qquad & \tau &= 13/20, \end{aligned} \tag{6}$$

and put $v = S - 2b > 0$. Here b is the band-shift parameter, not the word-width coefficient of Theorem 1.1.

4.1 Integer sizes and the switch comparison

Let $m = \lfloor \log_2 n \rfloor$ and take the even microblock size $s = 2\lfloor \lfloor m^{1/3} \rfloor / 2 \rfloor$. For $s > 0$, define

$$J = 20 \left\lfloor \frac{em}{20s} \right\rfloor, \qquad \mu = Js.$$

Then $20 \mid J$ and $0 \le em - \mu < 20s$, so $\mu/m \to e$. Use the integer band divisor $2^{\lfloor -b\mu \rfloor}$. It differs from the ideal real value by a factor below two. The divisibility of J ensures the terminal depth τJ is integral. Exceptional small sizes are handled by the finite fallback branch of the formal program.

Write $P(t) = \sum_{i=1}^{4} a_i t^i / Q$ with $Q = 10^8$, and $v = N/D$. At depth $j \le \tau J$, let $a = j/J$ and $r = (J - j)s > 0$. Select the least integer $q \in [0, r/2]$ satisfying

$$D(J-j) \sum_{i=1}^{4} a_i q^i r^{4-i} \;\ge\; Q(NJ + Dj) r^4. \tag{7}$$

All denominators are positive, so this comparison is exactly equivalent to $P(q/r) \ge (v+a)/(1-a)$. It uses only integer arithmetic. The certificate checks $P(1/2) > (v + \tau)/(1 - \tau)$; since r is even, an admissible q exists at every such depth. Minimality of q, together with $P' < 9$, gives

$$\frac{v+a}{1-a} \le P(q/r) < \frac{v+a}{1-a} + \frac{9}{r}.$$

With $t = q/r$, this implies

$$S \le 2b - a + (1-a)\gamma_{232}(t), \tag{8}$$

$$(1-a)H(t) < \rho(1+v) + 9\rho/\mu. \tag{9}$$

The certificate also checks $P(\theta) > v$. The initial integer threshold is at most $\theta + 1/\mu$ for the large-size branch. Its entropy is bounded using the tangent to H at θ. These facts provide uniform rounding control. The machine proof additionally accounts for scalar bounds, memory layout, and the execution cost of these comparisons.

4.2 Why the expansion can be 232

For $s = 2h > 0$ and $L = cs$, let

$$M = \binom{L}{s}, \quad V = \binom{L}{h}, \quad \Delta = \binom{L-h}{h}, \quad G = \binom{s}{h}.$$

Double counting disjoint ordered h-subsets gives $V\Delta = MG$. Since $\Delta \le V$ and $G < 2^s$, we obtain $\Delta^2 < M2^s$. Choosing one element from each of s disjoint c-element sets also gives $M \ge c^s > 2^s$. A prime selected above $\lceil\sqrt{M2^s}\rceil$ and within twice that integer satisfies $M2^s \le p^2 \le 16M2^s$, hence $\Delta < p$ and $G < p$. These finite inequalities do not require c to be a power of two.

Standard binomial entropy estimates yield

$$\log_2 M = cH(1/c)s + O(\log s), \qquad \log_2 V = cH(1/(2c))s + O(\log s),$$

and $\log_2 p = (E_c + 1/2)s + O(\log s)$. Moreover, $s^2/m \to 0$ and $J\log(s)/m \to 0$, so the associated local enumeration and polynomial factors are $n^{o(1)}$. These estimates alone do not certify the whole program: the concrete dimensions, bucket construction, reporting routines, and resource contracts must all use the same expansion. The accompanying extension supplies that machine-level connection.

5 The exact twelve-term certificate

The cost ledger retained from the baseline has the following exponent expressions, evaluated at $c = 232$ and the parameters above:

Term	Exponent expression
1	$2eE_c$
2	$1 + e(E_c - b)$
3	$2 - eS$
4	$2 - e(1/2 + b - H(\theta))$
5	$2 - e[1/2 - b - k_c - \rho(1 + S - 2b)]$
6	$2 - e(\tau - 1/2 - b - k_c)$
7	$\frac{8}{5}e(E_c + 1/2)$
8	$2 + e(4/5 - \frac{2}{5}E_c - 2b)$
9	$1 + e(1/2 - b)$
10	$2 - e(E_c + b)$
11	$2 - 2e(E_c + b)$
12	1

Exact interval evaluation proves

$$4.648781151243719 < E_{232} < 4.648781151243720,$$
$$0.001557987216578 < k_{232} < 0.001557987216579,$$
$$H(\theta) < 0.244168414687344.$$

The largest certified upper endpoint is

$$1.960613879576859694152281 6 < 1.96061388 < 1.960614.$$

The gap is positive and therefore absorbs the proved $n^{o(1)}$ overhead for sufficiently large inputs; finite exceptions are handled by the formal fallback. When E_c appears with a negative coefficient, its lower endpoint must be used to obtain an upper cost bound. The checker enforces this sign-sensitive interval calculation.

For completeness, the archived analytic package also includes degree-eight certificates at $c = 232$ and $c = 256$. The formal endpoint presented here uses the degree-four $c = 232$ choice. A separate fixed-c, uniform-entropy cost-model argument gives

$$1.96061349120 < \inf R_{\mathrm{model}}(232) < 1.960613492.$$

Its admissible parameters and proof are defined in Section 7 of the archived notes:

`analytic-certificate/PROOF.md`.

It is independent of the algorithmic endpoint and is not a lower bound for arbitrary 3SUM algorithms. It does not exclude changing the model, mixing expansions, or using different algorithmic constructions.

6 Connection to the complete formal program

The verified source includes the concrete expansion-232 cover and bucket construction, exact schedules, stored reporting data, host execution and resource bounds, and the final hybrid solver. In particular, the last theorem is not obtained by treating the numerical ledger as a machine runtime theorem. `FinalSolver.lean` invokes `ThreeSumHybridEndBridge.endStatement_of_solvers`, supplying the actual hybrid program, its correctness proof, and its polynomial resource bound for every magnitude exponent. The compiler bridge then yields the original word-RAM statement. The source is the authoritative detailed proof of this composition.

The archived official verifier receipt reports that `Race.bound` passed both the Lean and Nanoda kernels and that its statement matched the official frozen problem. Its reported axioms are exactly `propext`, `Classical.choice`, and `Quot.sound`, with no `sorryAx`. The receipt is dated October 14, 2026, and is tied to the source commit above and verifier commit `51489ee2ac82596a41375e92ea8108814f931242` [5].

The separately archived Prove2Me completion receipt, dated October 15, records completion of all seven original milestones, 41 published machine definition bundles, 481 accepted machine proof records, and 110 reused proved theorem nodes. The 481 records correspond to 480 new machine theorem declarations, not 481 independently novel mathematical results. A graph audit records 1,316 actual dependency paths. These are platform provenance statistics; they are not measures of novelty or substitutes for the standalone native source and verifier checks.

The official audit also records `site_eligible_for_record: false`. This report records successful verification without interpreting a leaderboard position as eligibility for a new speedrun record or as an independent assessment of scientific priority.

7 Reproduction and contents of the deposit

The source archive accompanying this report contains the pinned native proof source, the exact analytic certificates and their checker, selected verification evidence, original license notices, and a SHA-256 manifest. It excludes working-directory caches, credentials, unrelated research, and the incomplete later exponent-improvement work.

The analytic checker needs Python 3.10 or newer and only the standard library. From the extracted archive, run

```
python verify_release.py
python analytic-certificate/verify_certificate.py --self-test
```

The first command checks archived bytes and the official Lean-source hash map. It is an integrity check, not a new theorem proof. The second checks three exact certificates and twelve deliberately corrupted negative cases. The negative cases test rejection of wrong entropy ratios, oversized polynomial coefficients, overly small cost caps, and overly large model lower bounds.

Logarithms are enclosed using rational arithmetic. Reduce a positive rational argument to $y \in [1, 2]$ by powers of two and put $z = (y - 1)/(y + 1)$. For $K = 48$,

$$S_K \le \ln y \le S_K + \frac{2z^{2K+1}}{(2K+1)(1-z^2)}, \qquad S_K = 2\sum_{i=0}^{K-1} \frac{z^{2i+1}}{2i+1}.$$

The omitted series has positive terms, and the displayed geometric bound controls its remainder. Interval arithmetic accounts for negative range reduction exponents. The continuous lemmas in this report explain the logical use of the resulting finite inequalities.

Full formal reproduction uses Lean `v4.33.1`, with Mathlib pinned to

`0df444a360eaa60ab8c11dca51a86af692955474`.

It uses the pinned official verifier environment. The source follows the verifier's submission layout; the verifier supplies the Lake configuration. A root-directory build of the broader research workspace is not a reproduction of this theorem. The deposit preserves the dependency manifest and official logs. Preparing this deposit does not constitute a fresh full Lean or Nanoda run; the kernel conclusions above are attributed to the archived official run.

The original analytic notes retain their October 14 statement that the machine bridge was not yet established by that certificate package. Those historical files are preserved byte for byte. The subsequent native machine proof and official records are separate evidence and must not be confused with what the arithmetic checker itself proves.

8 Attribution and use of AI

The author is responsible for the submitted record. OpenAI Codex assisted the extension and formalization, inspection and organization of the artifacts, and preparation of this report. Automated checks and platform acceptance are not described as independent human peer review. This report does not assert that a separate human line-by-line audit has been completed.

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References

[1] Josh Alman and Virginia Vassilevska Williams. *Truly Subquadratic 3SUM and Truly Subcubic APSP via Triangles in Sparse Lopsided Graphs.* arXiv:2610.06783v1, 2026. `https://arxiv.org/abs/2610.06783v1`.

[2] c0ldzy17. *three-sum-lean*, pinned source and METHOD.md, commit `2b639655e07421b9ac743de9090459bab473a7c0`, 2026. `https://github.com/c0ldzy17/three-sum-lean/tree/2b639655e07421b9ac743de9090459bab473a7c0`.

[3] Anthropic. *Truly Subquadratic 3SUM and Truly Subcubic APSP: a Lean 4 formalization.* Source lineage and copyright notices are preserved in the accompanying `proof-source/NOTICE` and `proof-source/Submission/` tree.

[4] Yuxuan Xu (Anchen0823). *three-sum-1960614*, commit 8c25c0325de5be4286413f39ca0c9278aad33f52, 2026. https://github.com/Anchen0823/three-sum-1960614/tree/8c25c0325de5be4286413f39ca0c9278aad33f52.

[5] Big O Speedrun. Official verification issue and run for the preceding source, October 14, 2026. https://github.com/bigospeedrun/verifier/issues/8; https://github.com/bigospeedrun/verifier/actions/runs/38567645711.

[6] Prove2Me. *Deterministic Integer 3SUM 1.960614: Quartic Switch and Complete Program Extension*. Completion receipt archived October 15, 2026; campaign https://prove2.me/campaigns/3sum-exponent.